ESCOLA D'ENGINYERIA – UNIVERSITAT AUTÒNOMA DE BARCELONA

Practical Projects

FeedForward Neural Networks

REFERENCE TIME – SEQ + OpenMP

October 2026

Feed Forward Neural Networks

This document collects the reference times in sequential and parallel execution with OpenMP corresponding to the practice of parallelization of the neural network training code. In this activity, the students' goal is to analyze the provided code and implement their own parallel version using OpenMP, evaluating its performance.

The executions shown in this document have been carried out in:

- Wilma cluster, on nodes nodo.q and new-nodo.q
- Aolin Cluster, in the aolin.q node

The results presented serve as indicative reference values, so that each student can compare the behavior of their code with a functional and optimized implementation.

From the execution times, the speedup and efficiency values are also displayed, which allow you to analyze how the program scales when the number of threads increases.

Wilma (nodo.q)	
REFERENCE TIME	
Version	**Nº Epoch=10**
Sequential	40.737795

Version	Wilma (nodo.q)
OpenMP	REFERENCE TIME
Nº Threads	**Nº Epoch=10**
2	22.542747
4	11.730357
6	8.308845
12	5.263955

Version	Wilma (nodo.q)
OpenMP	SPEEDUP
Nº Threads	**Nº Epoch=10**
2	1.8071354
4	3.4728521
6	4.9029432
12	7.7390090

Version	Wilma (nodo.q)
OpenMP	EFICIENCIA
Nº Threads	Nº Epoch=10
2	0.9035677
4	0.8682130
6	0.8171572
12	0.6449174

Wilma (new-nodo.q)	
REFERENCE TIME	
Version	Nº Epoch=10
Sequential	12.095956

Version	Wilma (new-nodo.q)
OpenMP	REFERENCE TIME
Nº Threads	Nº Epoch=10
2	6.873421
4	4.296542
6	3.474935
12	2.555904

Version	Wilma (new-nodo.q)
OpenMP	SPEEDUP
Nº Threads	Nº Epoch=10
2	1.7598160
4	2.8152770
6	3.4809158
12	4.7325549

Version	Wilma (new-nodo.q)
OpenMP	EFFICIENCY
Nº Threads	Nº Epoch=10
2	0.8799080
4	0.7038193
6	0.5801526
12	0.3943796

Aolin (aolin.q)	
REFERENCE TIME	
Version	**Nº Epoch=10**
Sequential	11.23567

Version	Aolin (aolin.q)
OpenMP	REFERENCE TIME
Nº Threads	Nº Epoch=10
2	6.767024
4	3.65849
6	2.502528
12	2.380782

Version	Aolin (aolin.q)
OpenMP	SPEEDUP
Nº Threads	Nº Epoch=10
2	1.6603562
4	3.0711222
6	4.4897280
12	4.7193191

Version	Aolin (aolin.q)
OpenMP	EFFICIENCY
Nº Threads	Nº Epoch=10
2	0.8301781
4	0.7677806
6	0.7482880
12	0.3932766